TAURUS BIKE SHOP

Project Files Library

Version 4.0

AdminSystem (WPF / .NET 4.8) · WebApplication (ASP.NET Core MVC / .NET 8)

16 Flowcharts · 38 DB Tables · Schema v7.1

Architecture Ground Rules

AdminSystem	WPF desktop app (.NET 4.8). ALL admin logic lives here: POS, inventory, order management, logistics, reports, vouchers, support. Connects directly to TaurusBikeShopDB via ADO.NET/Dapper through App.config.
WebApplication	ASP.NET Core MVC (.NET 8). Customer-facing ONLY: product browsing, cart, checkout, payment, order tracking, reviews, wishlist, support ticket creation. Zero admin logic. EF Core via AppDbContext.
Shared Database	Both applications read/write TaurusBikeShopDB. They do NOT share code — only the database is shared.
COD Policy	Cash on Delivery is NOT offered. PaymentMethod: GCash | BankTransfer | Cash (POS only). No card payment.
Couriers	Lalamove and LBC only. No other courier values anywhere in code.
Stock Rule	Stock lives in ProductVariant.StockQuantity only (v7.0). Every Product must have >= 1 variant row.

SECTION 1 — File & Script Details

WEBAPPLICATION — Configuration & Root

WebApplication.slnx

Purpose	Visual Studio solution descriptor — canonical entry point for the WebApplication project.
Dependencies	WebApplication.csproj, Program.cs
Dev Notes	Always open via this file, not OOP-TaurusBikeShop.sln. net8.0 target.

WebApplication.csproj

Purpose	MSBuild project file — defines net8.0 target, EF Core NuGet packages, Razor compile settings.
Dependencies	Program.cs, appsettings.json, all .cs and .cshtml files
Dev Notes	Ensure TargetFramework is net8.0 only. Remove stale net7.0/net10.0 entries if they appear.

Program.cs

Purpose	Application entry point — configures DI container, middleware, EF Core, session, authentication, and all 5 background job registrations.
Dependencies	AppDbContext.cs, all IService interfaces, BackgroundJobs/, appsettings.json
Dev Notes	Register: AddHostedService for all 5 jobs. AddScoped for all services. AddDbContext. Configure auth middleware. Flowcharts: all 16 parts depend on this bootstrapping.

appsettings.json

Purpose	Production config — SQL Server connection string, logging levels, session settings.
Dependencies	AppDbContext.cs, Program.cs
Dev Notes	Connection string must point to TaurusBikeShopDB on the configured SQL Server host. Never commit dev secrets.

appsettings.Development.json

Purpose	Development overrides — local DB connection string, verbose logging.
Dependencies	appsettings.json, Program.cs
Dev Notes	Loaded when ASPNETCORE_ENVIRONMENT=Development. Can use a local DB instance.

WEBAPPLICATION — Background Jobs (Part 14)

InventorySyncJob.cs

Purpose	IHostedService — runs every 10 seconds. Compares ProductVariant.StockQuantity between POS and online channels and writes a unified InventoryLog entry.
Dependencies	ProductVariant entity, InventoryLog entity, AppDbContext, INotificationService. Flowchart: 14_BACKGROUND_JOBS.mmd Job 1.
Dev Notes	Implement as BackgroundService. Use CancellationToken in ExecuteAsync. Log every sync cycle to SystemLog.

PendingOrderMonitorJob.cs

Purpose	IHostedService — runs every 5 minutes. Finds Order rows with OrderStatus=Pending and CreatedAt older than 24 hours, then queues a PendingOrderAlert notification.

Dependencies	Order entity, Notification entity, AppDbContext, INotificationService. Flowchart: 14_BACKGROUND_JOBS.mmd Job 2.
Dev Notes	Only alert once per order — check if alert already sent. Write SystemLog entry on each run.

PaymentTimeoutJob.cs

Purpose	IHostedService — runs every 5 minutes. Finds BankTransferPayment rows where VerificationDeadline has passed and PaymentStatus is VerificationPending. Sets Order.OrderStatus to OnHold and queues PaymentHeld notification.
Dependencies	BankTransferPayment entity, Order entity, Notification entity, AppDbContext. Flowchart: 14_BACKGROUND_JOBS.mmd Job 3, 9_PAYMENT_VERIFICATION.mmd.
Dev Notes	Critical: use a DB transaction when updating both Payment and Order status atomically.

StockMonitorJob.cs

Purpose	IHostedService — runs every 15 minutes. Finds ProductVariant rows where StockQuantity is below ReorderThreshold. Queues LowStockAlert for admin and WishlistRestock notification for matching Wishlist customers.
Dependencies	ProductVariant entity, Wishlist entity, Notification entity, AppDbContext. Flowcharts: 14_BACKGROUND_JOBS.mmd Job 4, 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	Join Wishlist to ProductVariant to find affected customers. Avoid duplicate notifications per product per run.

DeliveryStatusPollJob.cs

Purpose	IHostedService — runs every 5 minutes. Polls Lalamove API for LalamoveDelivery rows and LBC API for LBCDelivery rows with non-final status. Updates Delivery.DeliveryStatus and queues TrackingUpdate notifications.
Dependencies	Delivery, LalamoveDelivery, LBCDelivery entities, Notification entity, AppDbContext, external courier APIs. Flowchart: 14_BACKGROUND_JOBS.mmd Job 5.
Dev Notes	Handle API failures gracefully — do not crash the job. Log API errors to SystemLog. Retry on next cycle.

WEBAPPLICATION — Business Logic: Interfaces

IBrandService.cs

Purpose	Interface contract for brand operations — GetAllBrands, GetById.
Dependencies	BrandService.cs, BrandRepository.cs, Brand entity. Used by ProductController.
Dev Notes	Keep minimal — brands are reference data for product filters. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.

ICartService.cs

Purpose	Interface contract for cart operations — AddItem, RemoveItem, UpdateQuantity, GetCart, ClearCart, MergeGuestCart.
Dependencies	CartService.cs, CartRepository.cs, Cart and CartItem entities, GuestSession entity. Flowcharts: 3_CART_CHECKOUT.mmd.
Dev Notes	Must support both authenticated and guest sessions. MergeGuestCart called when guest logs in.

INotificationService.cs

Purpose	Interface for queuing outbound Email/SMS notifications into the Notification table.
Dependencies	NotificationService.cs, Notification entity, AppDbContext. Referenced by all background jobs and most services. Flowcharts: all 16 parts.
Dev Notes	Queue only — does not dispatch. Background job dispatches. Method: Queue(channel, type, recipient, subject, body, userId, orderId?, ticketId?).

IOrderService.cs

Purpose	Interface for order lifecycle — CreateOrder, GetOrderHistory, GetOrderById, UpdateStatus, CancelOrder.
Dependencies	OrderService.cs, OrderRepository.cs, Order and OrderItem entities. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 5_ORDER_CONFIRMATION_TRACKING.mmd.
Dev Notes	CreateOrder must be atomic: write Order, write OrderItems, lock inventory, clear cart in a single transaction.

IPaymentService.cs

Purpose	Interface for payment processing — InitiateGCash, InitiateBankTransfer, RecordCashPayment, GetPaymentStatus. NOTE: ProcessCard must NOT exist — card payment is not offered.
Dependencies	PaymentService.cs, PaymentRepository.cs, Payment, GCashPayment, BankTransferPayment entities. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 9_PAYMENT_VERIFICATION.mmd.
Dev Notes	GCash: calls external gateway, stores GcashTransactionId. BankTransfer: stores proof URL, sets VerificationPending. Cash: POS only.

IProductService.cs

Purpose	Interface for product operations — GetProducts, Search, Filter, GetById, GetFeatured, GetVariants.
Dependencies	ProductService.cs, ProductRepository.cs, Product, ProductVariant, ProductImage, Category, Brand entities. Flowcharts: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	GetProducts must support filter by category, brand, price range, and search text. Include variant stock checks.

IReviewService.cs

Purpose	Interface for review operations — SubmitReview, GetProductReviews, VerifyPurchase.
Dependencies	ReviewService.cs, ReviewRepository.cs, Review entity, Order entity. Flowchart: 6_DELIVERY_REVIEW.mmd.
Dev Notes	VerifyPurchase checks that userId has a Delivered order containing productId before allowing review submission.

ISupportService.cs

Purpose	Interface for support ticket operations — CreateTicket, GetTicketsByUser, GetTicketById.
Dependencies	SupportService.cs, SupportRepository.cs, SupportTicket entity. Flowcharts: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd, 6_DELIVERY_REVIEW.mmd, 13_SUPPORT_SYSTEM.mmd.
Dev Notes	Web-side only creates and reads tickets. Admin-side (AdminSystem) handles reply, task assignment, and resolution.

IUserService.cs

Purpose	Interface for customer account operations — Register, Login, Logout, GetProfile, UpdateProfile, ChangePassword, VerifyOTP.
Dependencies	UserService.cs, UserRepository.cs, User entity, OTPVerification entity, GuestSession entity. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Register: hash password, send OTP, create User only after OTP verified. Login: validate hash, update LastLoginAt.

IVoucherService.cs

Purpose	Interface for voucher operations — ValidateCode, ApplyVoucher, GetDiscountAmount.
Dependencies	VoucherService.cs, VoucherRepository.cs, Voucher, UserVoucher, VoucherUsage entities. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	ValidateCode performs all 5 checks: exists+active, date range, MinimumOrderAmount, MaxUses, MaxUsesPerUser. Returns typed error on failure.

IWishlistService.cs

Purpose	Interface for wishlist operations — AddToWishlist, RemoveFromWishlist, GetWishlist, IsInWishlist.

Dependencies	WishlistService.cs, WishlistRepository.cs, Wishlist entity. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	AddToWishlist is idempotent — ignore duplicate constraint. GetWishlist returns product details joined.

WEBAPPLICATION — Business Logic: Services

BrandService.cs

Purpose	Implements IBrandService — retrieves brand list for product filters and detail pages.
Dependencies	IBrandService.cs, BrandRepository.cs, Brand entity.
Dev Notes	Simple read-only service. Cache results if performance is a concern.

CartService.cs

Purpose	Implements ICartService — full cart lifecycle including guest cart support and merge on login.
Dependencies	ICartService.cs, CartRepository.cs, Cart and CartItem entities, GuestSession entity. Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	Active cart = IsCheckedOut=false. One active cart per user (unique index). On login, merge guest cart items into user cart.

NotificationService.cs

Purpose	Implements INotificationService — writes Notification rows to the database queue.
Dependencies	INotificationService.cs, Notification entity, AppDbContext. Flowcharts: all 16 parts.
Dev Notes	Does not dispatch email/SMS directly — only inserts the row. Dispatch handled by background polling logic (vw_PendingNotifications filters RetryCount < 3).

OrderService.cs

Purpose	Implements IOrderService — creates orders from cart, manages status, links payment and delivery.
Dependencies	IOrderService.cs, OrderRepository.cs, Order, OrderItem, InventoryLog entities, INotificationService. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 5_ORDER_CONFIRMATION_TRACKING.mmd.
Dev Notes	CreateOrder: copy cart items to OrderItems, write InventoryLog Lock entries, clear cart, set OrderStatus=Pending. All in one DB transaction.

PaymentService.cs

Purpose	Implements IPaymentService — GCash integration, bank transfer proof upload, cash recording.
Dependencies	IPaymentService.cs, PaymentRepository.cs, Payment, GCashPayment, BankTransferPayment entities, INotificationService. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 9_PAYMENT_VERIFICATION.mmd.
Dev Notes	BankTransfer: store proof in GCS, write ProofUrl/Bucket/Path to BankTransferPayment, set VerificationDeadline = now+24h.

ProductService.cs

Purpose	Implements IProductService — product listing, search, filter, variant and image retrieval.
Dependencies	IProductService.cs, ProductRepository.cs, Product, ProductVariant, ProductImage, Category, Brand entities. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	GetProducts: filter by CategoryId, BrandId, price range, search string. Return includes primary image URL and first variant stock.

ReviewService.cs

Purpose	Implements IReviewService — review submission with verified-purchase check.
Dependencies	IReviewService.cs, ReviewRepository.cs, Review entity, Order entity. Flowchart: 6_DELIVERY_REVIEW.mmd.

Dev Notes	Before insert: verify userId has a Delivered order containing productId. Set IsVerifiedPurchase=true if confirmed.

SupportService.cs

Purpose	Implements ISupportService — customer-facing ticket creation and read.
Dependencies	ISupportService.cs, SupportRepository.cs, SupportTicket entity, INotificationService. Flowcharts: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd, 6_DELIVERY_REVIEW.mmd.
Dev Notes	CreateTicket: write SupportTicket row, queue SupportTicketCreated notification to customer and admin.

UserService.cs

Purpose	Implements IUserService — registration, OTP flow, login, profile update.
Dependencies	IUserService.cs, UserRepository.cs, User, OTPVerification, Address entities, PasswordHelper, INotificationService. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Register: insert OTPVerification, send OTPCode notification. VerifyOTP: flip IsUsed, create User + Address. Login: bcrypt verify, set LastLoginAt.

VoucherService.cs

Purpose	Implements IVoucherService — 5-step voucher validation chain and discount calculation.
Dependencies	IVoucherService.cs, VoucherRepository.cs, Voucher, UserVoucher, VoucherUsage entities. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	Validate in order: IsActive+exists, date range, MinimumOrderAmount, MaxUses count from VoucherUsage, per-user count. Return DiscountAmount as Percentage or Fixed.

WishlistService.cs

Purpose	Implements IWishlistService — add/remove/list saved products per customer.
Dependencies	IWishlistService.cs, WishlistRepository.cs, Wishlist entity, Product entity. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	AddToWishlist: ignore UniqueConstraint violation (idempotent). GetWishlist: join to Product for display info.

WEBAPPLICATION — Controllers

CartController.cs

Purpose	Handles cart AJAX endpoints — AddToCart, RemoveFromCart, UpdateQuantity, ViewCart, GetCartCount.
Dependencies	ICartService, Cart.cshtml, CartViewModel, _CartItem.cshtml. Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	AddToCart and Remove are AJAX (return JSON). ViewCart returns full Cart.cshtml. Support both userId and GuestSessionId. Flowchart: 3_CART_CHECKOUT.mmd.

CheckoutController.cs

Purpose	Handles multi-step checkout: review cart, enter/confirm address, select delivery, apply voucher, create order.
Dependencies	ICartService, IOrderService, IVoucherService, Checkout.cshtml, CheckoutViewModel, _OrderSummary.cshtml, _DeliverySelection.cshtml. Flowcharts: 3_CART_CHECKOUT.mmd, 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	Orchestrates the cart-to-order transition. Voucher validation done via AJAX call to VoucherController. Order creation must be atomic.

CustomerController.cs

Purpose	Handles customer auth routes — Register, Login, Logout, Dashboard, Profile, EditProfile.
Dependencies	IUserService, Login.cshtml, Register.cshtml, Dashboard.cshtml, Profile.cshtml, LoginViewModel, RegisterViewModel, ProfileViewModel. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.

Dev Notes	Session management: set userId in session on login, clear on logout. OTP flow: Register -> POST -> VerifyOTP -> create account.

HomeController.cs

Purpose	Handles landing page and privacy page with featured product display.
Dependencies	IProductService, Index.cshtml, Privacy.cshtml, ProductViewModel.
Dev Notes	Load IsFeatured=true products for homepage hero section. No authentication required.

OrderController.cs

Purpose	Handles customer order views — OrderHistory, OrderDetail, TrackOrder.
Dependencies	IOrderService, OrderHistory.cshtml, Confirmation.cshtml, OrderViewModel. Flowcharts: 5_ORDER_CONFIRMATION_TRACKING.mmd.
Dev Notes	TrackOrder shows real-time status from Order + Delivery tables. OrderHistory paginates by userId.

PaymentController.cs

Purpose	Handles payment submission, GCash callback, bank transfer proof upload.
Dependencies	IPaymentService, INotificationService, Payment.cshtml, _PaymentMethod.cshtml, _OTPModal.cshtml. Flowcharts: 4_PAYMENT_PROCESSING.mmd.
Dev Notes	GCash: redirect to gateway, handle callback, update PaymentStatus. BankTransfer: accept proof file upload, store to GCS, set VerificationPending.

ProductController.cs

Purpose	Handles product catalogue — list with filters, search, and product detail page.
Dependencies	IProductService, IBrandService, ProductCatalog.cshtml, ProductDetails.cshtml, ProductViewModel, ProductDetailViewModel. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	List: query string params for category, brand, price range, search, page. Detail: load all variants, images, and reviews.

ReviewController.cs

Purpose	Handles review submission (POST) and retrieval (AJAX for product detail).
Dependencies	IReviewService, Reviews.cshtml, ReviewViewModel. Flowchart: 6_DELIVERY_REVIEW.mmd.
Dev Notes	POST SubmitReview: verify purchase before insert. GET GetProductReviews: returns JSON list for AJAX load on product detail.

SupportController.cs

Purpose	Handles support ticket creation from customer-facing pages.
Dependencies	ISupportService, INotificationService. Flowcharts: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd, 6_DELIVERY_REVIEW.mmd, 13_SUPPORT_SYSTEM.mmd.
Dev Notes	POST Create: receives category, subject, description, optional orderId. Writes SupportTicket, queues notifications to customer and admin.

VoucherController.cs

Purpose	Handles voucher validation AJAX endpoint — ValidateCode returns discount amount or error.
Dependencies	IVoucherService, Voucher entity. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	Returns JSON: { valid, discountType, discountAmount, errorMessage }. Called by checkout.js on code entry. No authentication check — server-side usage limit enforced.

WishlistController.cs

Purpose	Handles wishlist toggle (AJAX AddToWishlist, RemoveFromWishlist) and ViewWishlist page.
Dependencies	IWishlistService, Wishlist.cshtml, WishlistViewModel. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Add/Remove are AJAX — return updated wishlist count for badge update. ViewWishlist loads full Wishlist.cshtml.

WEBAPPLICATION — Data Access

AppDbContext.cs

Purpose	EF Core DbContext — defines all 38 DbSet<T> entities, configures 1:1 relationships, indexes, and schema constraints.
Dependencies	All 38 entity files in Models/Entities/, appsettings.json (connection string), Program.cs. Schema: Taurus_schema_v7.1.sql.
Dev Notes	Configure 1:1 in OnModelCreating: GCashPayment, BankTransferPayment, LalamoveDelivery, LBCDelivery, PickupOrder all share PK with their parent. Add HasOne/WithOne/HasForeignKey chains.

IRepository.cs

Purpose	Generic repository interface — GetAll, GetById, Add, Update, Delete, Find.
Dependencies	Repository.cs, all specific repositories, AppDbContext.
Dev Notes	Base contract for all data access. Type parameter T is the entity class.

Repository.cs

Purpose	Generic EF Core repository implementation — provides default CRUD using AppDbContext.
Dependencies	IRepository.cs, AppDbContext.cs.
Dev Notes	Specific repos inherit this and add domain-specific query methods.

BrandRepository.cs

Purpose	Brand CRUD — GetAllBrands, GetById, used in product filter dropdowns.
Dependencies	Repository.cs, Brand entity, AppDbContext.
Dev Notes	Read-only in production. Brands added via AdminSystem.

CartRepository.cs

Purpose	Cart and CartItem CRUD — create cart, add/remove items, get active cart for user or guest.
Dependencies	Repository.cs, Cart and CartItem entities, AppDbContext. Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	GetActiveCart: filter IsCheckedOut=false by userId or guestSessionId. Enforce one active cart per user via index.

OrderRepository.cs

Purpose	Order and OrderItem CRUD — create order from checkout, get history by userId, update status.
Dependencies	Repository.cs, Order and OrderItem entities, AppDbContext. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 5_ORDER_CONFIRMATION_TRACKING.mmd.
Dev Notes	CreateOrder must wrap Order + OrderItem inserts in a transaction. GetOrderById: include OrderItems with Product details.

PaymentRepository.cs

Purpose	Payment, GCashPayment, BankTransferPayment CRUD.
Dependencies	Repository.cs, Payment, GCashPayment, BankTransferPayment entities, AppDbContext. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 9_PAYMENT_VERIFICATION.mmd.

Dev Notes	After inserting Payment, always insert the corresponding subtype row (GCashPayment or BankTransferPayment) in the same call.

ProductRepository.cs

Purpose	Product queries — GetAll with filtering/search, GetById with full joins, stock level checks.
Dependencies	Repository.cs, Product, ProductVariant, ProductImage, Category, Brand entities, AppDbContext. Flowcharts: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	GetAll: build dynamic EF query from filter params. GetById: include Variants, Images, Reviews. Apply IsActive=true filter by default.

ReviewRepository.cs

Purpose	Review CRUD — submit review, get reviews by product, check verified purchase.
Dependencies	Repository.cs, Review entity, Order entity, AppDbContext. Flowchart: 6_DELIVERY_REVIEW.mmd.
Dev Notes	HasVerifiedPurchase(userId, productId): check if Order with Delivered status contains the productId for that user.

SupplierRepository.cs

Purpose	Supplier and PurchaseOrder CRUD — create/read/update supplier records and purchase orders.
Dependencies	Repository.cs, Supplier, PurchaseOrder, PurchaseOrderItem, ProductVariant entities, AppDbContext. Flowchart: 16_SUPPLIER_PURCHASE_ORDER.mmd.
Dev Notes	ReceiveStock: update ProductVariant.StockQuantity and write InventoryLog Purchase entries per item. Used by AdminSystem only via shared DB.

SupportRepository.cs

Purpose	SupportTicket CRUD — create ticket, get by user, get by id with replies.
Dependencies	Repository.cs, SupportTicket, SupportTicketReply, SupportTask entities, AppDbContext. Flowcharts: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd, 13_SUPPORT_SYSTEM.mmd.
Dev Notes	GetTicketById: include Replies ordered by CreatedAt. Web side only creates and reads; admin side writes replies and tasks.

UserRepository.cs

Purpose	User CRUD — register, find by email (login), update profile, get by ID.
Dependencies	Repository.cs, User, Address, OTPVerification, GuestSession entities, AppDbContext. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	FindByEmail: used for login. CreateOTP: insert OTPVerification row. VerifyOTP: set IsUsed=true, return success. Check ExpiresAt.

VoucherRepository.cs

Purpose	Voucher lookup, usage count check, write VoucherUsage on redemption.
Dependencies	Repository.cs, Voucher, UserVoucher, VoucherUsage entities, AppDbContext. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	GetUsageCount(voucherId): COUNT(*) from VoucherUsage. GetUserUsageCount(voucherId, userId): per-user count. WriteUsage: insert VoucherUsage row.

WishlistRepository.cs

Purpose	Wishlist add/remove/list by userId.
Dependencies	Repository.cs, Wishlist entity, Product entity, AppDbContext. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Add: catch duplicate key silently (idempotent). GetByUser: join to Product for display. GetProductIdsInWishlist: used by stock monitor job.

WEBAPPLICATION — Models: Entities (38 Tables)

Address.cs

Purpose	EF Core entity for Address table — user shipping addresses and order snapshot copies.
Dependencies	User entity, Order entity, AppDbContext. Flowcharts: 3_CART_CHECKOUT.mmd.
Dev Notes	IsSnapshot=true rows are created at checkout and never edited. IsDefault=true marks user primary address.

BankTransferPayment.cs

Purpose	EF Core entity for BankTransferPayment table — 1:1 with Payment (shared PK).
Dependencies	Payment entity, User entity, AppDbContext. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 9_PAYMENT_VERIFICATION.mmd.
Dev Notes	VerificationDeadline: set to now+24h on upload. Polled by PaymentTimeoutJob. ProofUrl: GCS public URL.

Brand.cs

Purpose	EF Core entity for Brand table.
Dependencies	Product entity, AppDbContext. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Read-only from web. Managed by admin.

Cart.cs

Purpose	EF Core entity for Cart table — parent record for a shopping session.
Dependencies	CartItem entity, User entity, GuestSession entity, AppDbContext. Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	CHECK constraint: exactly one of UserId or GuestSessionId must be set. IsCheckedOut flips to true on order creation.

CartItem.cs

Purpose	EF Core entity for CartItem table — individual product lines in a cart.
Dependencies	Cart entity, Product entity, ProductVariant entity, AppDbContext. Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	PriceAtAdd: snapshot of price when added. Quantity CHECK > 0.

Category.cs

Purpose	EF Core entity for Category table — self-referencing hierarchy.
Dependencies	Product entity, AppDbContext. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	ParentCategoryId is nullable self-ref. CategoryCode is unique (UNIT, FRAME, FORK, etc.).

Delivery.cs

Purpose	EF Core entity for Delivery table — base delivery record, courier-agnostic.
Dependencies	Order entity, LalamoveDelivery entity, LBCDelivery entity, AppDbContext. Flowcharts: 5_ORDER_CONFIRMATION_TRACKING.mmd, 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	Courier CHECK: Lalamove or LBC only. IsDelayed + DelayedUntil set by admin in Part 10.

GCashPayment.cs

Purpose	EF Core entity for GCashPayment table — 1:1 with Payment (shared PK).
Dependencies	Payment entity, AppDbContext. Flowchart: 4_PAYMENT_PROCESSING.mmd.
Dev Notes	GcashTransactionId populated after gateway callback. NULL while pending.

GuestSession.cs

Purpose	EF Core entity for GuestSession table — anonymous shopping session.
Dependencies	Cart entity, User entity, AppDbContext. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	SessionToken: client-generated UUID stored in cookie. ConvertedToUserId set when guest registers.

InventoryLog.cs

Purpose	EF Core entity for InventoryLog table — immutable stock movement audit trail.
Dependencies	Product entity, ProductVariant entity, Order entity, PurchaseOrder entity, User entity, AppDbContext. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 6_DELIVERY_REVIEW.mmd, 12_INVENTORY_MANAGEMENT.mmd, 16_SUPPLIER_PURCHASE_ORDER.mmd.
Dev Notes	ChangeType CHECK: Purchase, Sale, Return, Adjustment, Damage, Loss, Lock, Unlock. Never delete or update rows.

LalamoveDelivery.cs

Purpose	EF Core entity for LalamoveDelivery table — 1:1 with Delivery (shared PK).
Dependencies	Delivery entity, AppDbContext. Flowcharts: 5_ORDER_CONFIRMATION_TRACKING.mmd, 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	BookingRef, DriverName, DriverPhone all NULL until Lalamove confirms booking and assigns driver.

LBCDelivery.cs

Purpose	EF Core entity for LBCDelivery table — 1:1 with Delivery (shared PK).
Dependencies	Delivery entity, AppDbContext. Flowcharts: 5_ORDER_CONFIRMATION_TRACKING.mmd, 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	TrackingNumber NULL until LBC waybill generated. No driver columns — LBC does not expose driver info via API.

Notification.cs

Purpose	EF Core entity for Notification table — outbound Email/SMS queue.
Dependencies	User entity, Order entity, SupportTicket entity, AppDbContext. Flowcharts: referenced in all 16 parts.
Dev Notes	Status CHECK: Pending, Sent, Failed. RetryCount: background job stops at 3. Channel CHECK: Email, SMS.

Order.cs

Purpose	EF Core entity for Order table — core order record.
Dependencies	User entity, Address entity, OrderItem entity, Payment entity, Delivery entity, PickupOrder entity, AppDbContext. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 5_ORDER_CONFIRMATION_TRACKING.mmd, 8_ONLINE_ORDERS_MANAGEMENT.mmd.
Dev Notes	TotalAmount not stored — computed in vw_OrderSummary as SubTotal-DiscountAmount+ShippingFee. OrderStatus CHECK: 9 values.

OrderItem.cs

Purpose	EF Core entity for OrderItem table — line items locked at order time.
Dependencies	Order entity, Product entity, ProductVariant entity, AppDbContext.
Dev Notes	UnitPrice and Quantity are immutable after creation. Subtotal computed in vw_OrderItemDetail.

OTPVerification.cs

Purpose	EF Core entity for OTPVerification table — registration OTP codes.
Dependencies	AppDbContext. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	No FK to User — created before account exists. ExpiresAt typically now+10min. IsUsed flips to true after verification.

Payment.cs

Purpose	EF Core entity for Payment table — method-agnostic base payment record.
Dependencies	Order entity, GCashPayment entity, BankTransferPayment entity, AppDbContext. Flowcharts: 4_PAYMENT_PROCESSING.mmd, 9_PAYMENT_VERIFICATION.mmd.
Dev Notes	PaymentMethod CHECK: GCash, BankTransfer, Cash. No Card. PaymentStatus CHECK: 6 values including VerificationPending and VerificationRejected.

PickupOrder.cs

Purpose	EF Core entity for PickupOrder table — 1:1 with Order for store pickup.
Dependencies	Order entity, AppDbContext. Flowcharts: 3_CART_CHECKOUT.mmd, 8_ONLINE_ORDERS_MANAGEMENT.mmd.
Dev Notes	Only exists for pickup orders. PickupExpiresAt = PickupReadyAt + 3 days. Polled by PendingOrderMonitorJob.

POS_Session.cs

Purpose	EF Core entity for POS_Session table — walk-in terminal session.
Dependencies	User entity, AppDbContext. Flowchart: 11_WALKIN_POS.mmd.
Dev Notes	ShiftEnd NULL while session is active. TotalSales running total for the shift.

PriceHistory.cs

Purpose	EF Core entity for PriceHistory table — auto-populated by TR_Product_PriceAudit trigger.
Dependencies	Product entity, User entity, AppDbContext.
Dev Notes	Read-only from application. Written by DB trigger on Product.Price change. ChangedByUserId NULL when trigger-generated.

Product.cs

Purpose	EF Core entity for Product table — core product master.
Dependencies	Category entity, Brand entity, ProductVariant entity, ProductImage entity, PriceHistory entity, AppDbContext. Flowcharts: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	No StockQuantity column — stock lives in ProductVariant only (v7.0 fix). Every product must have >= 1 variant.

ProductImage.cs

Purpose	EF Core entity for ProductImage table — GCS image metadata.
Dependencies	Product entity, User entity, AppDbContext.
Dev Notes	Stores only URL and GCS path metadata. Binary in Google Cloud Storage. IsPrimary: only one per product (filtered unique index).

ProductVariant.cs

Purpose	EF Core entity for ProductVariant table — stock, pricing variants, and ReorderThreshold.
Dependencies	Product entity, CartItem entity, OrderItem entity, InventoryLog entity, AppDbContext. Flowcharts: 12_INVENTORY_MANAGEMENT.mmd, 14_BACKGROUND_JOBS.mmd.
Dev Notes	ReorderThreshold (v7.1): StockMonitorJob alerts when StockQuantity < ReorderThreshold. TotalPrice = Product.Price + AdditionalPrice.

PurchaseOrder.cs

Purpose	EF Core entity for PurchaseOrder table — supplier restocking orders.
Dependencies	Supplier entity, PurchaseOrderItem entity, User entity, AppDbContext. Flowchart: 16_SUPPLIER_PURCHASE_ORDER.mmd.
Dev Notes	Status CHECK: Pending, Received, Cancelled. TotalAmount not stored — computed in vw_PurchaseOrderDetail.

PurchaseOrderItem.cs

Purpose	EF Core entity for PurchaseOrderItem table — line items of a supplier PO.
Dependencies	PurchaseOrder entity, Product entity, ProductVariant entity, AppDbContext. Flowchart: 16_SUPPLIER_PURCHASE_ORDER.mmd.
Dev Notes	UnitPrice locked at PO time. Subtotal computed. Quantity CHECK > 0.

Review.cs

Purpose	EF Core entity for Review table.
Dependencies	User entity, Product entity, Order entity, AppDbContext. Flowchart: 6_DELIVERY_REVIEW.mmd.
Dev Notes	Rating CHECK: 1-5. IsVerifiedPurchase set by ReviewService after confirming Delivered order.

Role.cs

Purpose	EF Core entity for Role table — system roles.
Dependencies	UserRole entity, AppDbContext. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	RoleName CHECK: Admin, Manager, Cashier, Staff, Customer.

Supplier.cs

Purpose	EF Core entity for Supplier table.
Dependencies	PurchaseOrder entity, AppDbContext. Flowchart: 16_SUPPLIER_PURCHASE_ORDER.mmd.
Dev Notes	Email is unique when non-NULL (filtered index). IsActive flag.

SupportTask.cs

Purpose	EF Core entity for SupportTask table (v7.1) — actionable follow-up tasks on tickets.
Dependencies	SupportTicket entity, User entity, AppDbContext. Flowchart: 13_SUPPORT_SYSTEM.mmd.
Dev Notes	TaskType CHECK: ShipReplacement, ArrangeReturn, IssueRefund, ContactSupplier, Other. TaskStatus CHECK: Pending, InProgress, Done, Cancelled.

SupportTicket.cs

Purpose	EF Core entity for SupportTicket table.
Dependencies	User entity, Order entity, SupportTicketReply entity, SupportTask entity, Notification entity, AppDbContext. Flowcharts: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd, 6_DELIVERY_REVIEW.mmd, 13_SUPPORT_SYSTEM.mmd.
Dev Notes	TicketCategory CHECK: 7 values. TicketStatus CHECK: Open, InProgress, AwaitingResponse, Resolved, Closed. AssignedToUserId nullable.

SupportTicketReply.cs

Purpose	EF Core entity for SupportTicketReply table — reply thread per ticket.
Dependencies	SupportTicket entity, User entity, AppDbContext. Flowchart: 13_SUPPORT_SYSTEM.mmd.
Dev Notes	IsAdminReply=true for staff messages. CASCADE DELETE from SupportTicket.

SystemLog.cs

Purpose	EF Core entity for SystemLog table — admin action and background job audit trail.
Dependencies	User entity, AppDbContext. Flowchart: 14_BACKGROUND_JOBS.mmd.
Dev Notes	EventType CHECK: 22 values covering auth, product, order, payment, inventory, delivery, background job, and support events. UserId NULL for automated events.

User.cs

Purpose	EF Core entity for User table — core customer/staff account.
Dependencies	Address entity, UserRole entity, Cart entity, Order entity, Wishlist entity, Review entity, AppDbContext. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Email unique filtered index. IsWalkIn=true for POS placeholder. DefaultAddressId is deferred circular FK.

UserRole.cs

Purpose	EF Core entity for UserRole junction table — N:M between User and Role.
Dependencies	User entity, Role entity, AppDbContext. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	UNIQUE(UserId, RoleId). CASCADE DELETE from User.

UserVoucher.cs

Purpose	EF Core entity for UserVoucher table — assigns vouchers to specific users.
Dependencies	User entity, Voucher entity, AppDbContext. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	ExpiresAt overrides Voucher.EndDate for this user. UNIQUE(UserId, VoucherId).

Voucher.cs

Purpose	EF Core entity for Voucher table — discount voucher definitions.
Dependencies	UserVoucher entity, VoucherUsage entity, AppDbContext. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	DiscountType CHECK: Percentage or Fixed. MaxUses and MaxUsesPerUser are nullable (NULL = unlimited).

VoucherUsage.cs

Purpose	EF Core entity for VoucherUsage table — redemption log.
Dependencies	Voucher entity, User entity, Order entity, AppDbContext. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	One row per voucher redemption per order. Used to enforce MaxUses and MaxUsesPerUser counts.

Wishlist.cs

Purpose	EF Core entity for Wishlist table — customer saved products.
Dependencies	User entity, Product entity, AppDbContext. Flowcharts: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd, 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	UNIQUE(UserId, ProductId). StockMonitorJob reads this to send WishlistRestock notifications.

WEBAPPLICATION — Models: ViewModels

CartViewModel.cs

Purpose	Data shape for Cart.cshtml — CartItems list, subtotal, item count, grand total.
Dependencies	Cart.cshtml, CartController.cs, CartItem entity. Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	Include VoucherCode field for Checkout flow handoff. GrandTotal = SubTotal - Discount + Shipping.

CheckoutViewModel.cs

Purpose	Data shape for Checkout.cshtml — shipping address, delivery method, voucher code, order summary.
Dependencies	Checkout.cshtml, CheckoutController.cs, Order entity. Flowcharts: 3_CART_CHECKOUT.mmd, 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	Validation attributes on ShippingAddress fields. Include VoucherCode string and AppliedDiscountAmount decimal.

LoginViewModel.cs

Purpose	Data shape for Login.cshtml — Email, Password, RememberMe.
Dependencies	Login.cshtml, CustomerController.cs. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Required + EmailAddress data annotations. Password is [DataType.Password].

OrderViewModel.cs

Purpose	Data shape for OrderHistory.cshtml and Confirmation.cshtml.
Dependencies	OrderHistory.cshtml, Confirmation.cshtml, OrderController.cs, Order entity. Flowchart: 5_ORDER_CONFIRMATION_TRACKING.mmd.
Dev Notes	Include OrderNumber, Status badge text, formatted TotalAmount, ItemCount, DeliveryETA.

ProductDetailViewModel.cs

Purpose	Rich product detail model — specs, image gallery, variants, reviews, related products.
Dependencies	ProductDetails.cshtml, ProductController.cs, Product and ProductVariant entities. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	SelectedVariantId drives add-to-cart. ReviewList included for initial render. IsInWishlist boolean for heart icon.

ProductViewModel.cs

Purpose	Product card model for listing grid — Name, Price, SKU, Brand, PrimaryImageUrl, StockQuantity.
Dependencies	_ProductCard.cshtml, ProductCatalog.cshtml, ProductController.cs. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	StockQuantity should aggregate from variants. IsInWishlist included for wishlist toggle button state.

ProfileViewModel.cs

Purpose	Profile page model — display and edit personal info and addresses.
Dependencies	Profile.cshtml, CustomerController.cs, User and Address entities. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Include list of saved addresses. IsDefault flag drives address management UI.

RegisterViewModel.cs

Purpose	Registration form model — FirstName, LastName, Email, Password, ConfirmPassword, Phone, Address.
Dependencies	Register.cshtml, CustomerController.cs. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Compare attribute: Password == ConfirmPassword. Address fields map to Address entity at account creation.

ReviewViewModel.cs

Purpose	Review model — Rating, Comment, ProductId, ProductName, IsVerifiedPurchase.
Dependencies	Reviews.cshtml, ProductDetails.cshtml, ReviewController.cs. Flowchart: 6_DELIVERY_REVIEW.mmd.
Dev Notes	Rating 1-5 star UI widget driven by review.js. IsVerifiedPurchase returned from ReviewService for display badge.

WishlistViewModel.cs

Purpose	Wishlist page model — saved products with Name, Price, ImageUrl, InStock flag.
Dependencies	Wishlist.cshtml, WishlistController.cs, Wishlist entity. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	InStock checks ProductVariant.StockQuantity > 0. Include AddToCart CTA when in stock.

ErrorViewModel.cs

Purpose	Standard MVC error model — RequestId, ShowRequestId.
Dependencies	Error.cshtml, HomeController.cs.

Dev Notes	Auto-generated by ASP.NET Core template. Only used for unhandled exception fallback.

WEBAPPLICATION — Utilities

FileUploadHelper.cs

Purpose	Handles product image and payment proof uploads — saves files to GCS and returns URL + path metadata.
Dependencies	ProductController.cs (admin, via shared DB), PaymentController.cs, ProductImage entity, BankTransferPayment entity.
Dev Notes	Returns (ImageUrl, StorageBucket, StoragePath) tuple. Validate MIME type (image/webp, image/jpeg). Max file size check.

PasswordHelper.cs

Purpose	BCrypt password hashing and verification for customer account security.
Dependencies	UserService.cs, UserRepository.cs, User entity. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Hash(password): BCrypt.Net.BCrypt.HashPassword. Verify(password, hash): BCrypt.Net.BCrypt.Verify.

ValidationHelper.cs

Purpose	Custom validation — phone number format, postal code, email extra rules.
Dependencies	CustomerController.cs, CheckoutController.cs, ViewModels.
Dev Notes	Philippine phone format: +63 or 09 prefix. Postal code: 4-digit numeric. Used before DB writes.

WEBAPPLICATION — Views

_ViewImports.cshtml

Purpose	Global Razor directives — @using namespaces and @addTagHelper for all views.
Dependencies	All .cshtml files, _Layout.cshtml.
Dev Notes	Add @using for Models/Entities, Models/ViewModels namespaces. Eliminates repetitive @using in every view.

_ViewStart.cshtml

Purpose	Sets default layout: Layout = ~/Views/Shared/_Layout.cshtml for all views.
Dependencies	_Layout.cshtml, all Customer and Home views.
Dev Notes	Admin pages in AdminSystem do not use this — WPF only.

Index.cshtml

Purpose	Landing page — hero banner, featured products, category grid.
Dependencies	HomeController.cs, ProductViewModel, Index.css. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Load IsFeatured=true products. No authentication required. CTA to ProductCatalog.

Privacy.cshtml

Purpose	Static privacy policy page.
Dependencies	HomeController.cs, _Layout.cshtml.
Dev Notes	Legal compliance. No dynamic data.

Login.cshtml

Purpose	Login form — email, password, remember-me, link to Register.
Dependencies	CustomerController.cs, LoginViewModel.cs, auth.css. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Redirect to Dashboard on success. Show ReturnUrl for post-login redirect after attempted protected route.

Register.cshtml

Purpose	Registration form — name, email, phone, address, password, confirm.
Dependencies	CustomerController.cs, RegisterViewModel.cs, register.css, _OTPModal.cshtml. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	POST to Register action triggers OTP send. OTP modal (_OTPModal.cshtml) shown for verification step.

Dashboard.cshtml

Purpose	Customer dashboard — recent orders, wishlist count, profile summary, quick links.
Dependencies	CustomerController.cs, DashboardViewModel, dashboard.css. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Requires authentication. Display notification badges for unread alerts. Quick links to OrderHistory, Wishlist, Support.

Profile.cshtml

Purpose	Profile view/edit — personal info, saved addresses, change password.
Dependencies	CustomerController.cs, ProfileViewModel, profile.css. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Address list: set default, add new, delete. Password change: current password verification required.

ProductCatalog.cshtml

Purpose	Product listing grid — filter sidebar, search bar, sort, pagination.
Dependencies	ProductController.cs, ProductViewModel, _ProductCard.cshtml, product-catalog.css, product-catalog.js. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Filters (category, brand, price) via AJAX reload. Search input debounced in product-catalog.js. Wishlist toggle AJAX button on each card.

ProductDetails.cshtml

Purpose	Single product — image gallery, spec table, variant selector, add-to-cart, reviews.
Dependencies	ProductController.cs, ProductDetailViewModel, product-details.css, review.js. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Variant selection updates add-to-cart price dynamically. Review section loads via AJAX. Out-of-stock shows wishlist button.

Cart.cshtml

Purpose	Cart page — item list with qty steppers, remove, subtotal, checkout CTA.
Dependencies	CartController.cs, CartViewModel, _CartItem.cshtml, cart.css, cart.js. Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	Qty stepper updates via AJAX to cart.js. Remove item AJAX. Subtotal recalculates client-side. Checkout CTA to CheckoutController.

Checkout.cshtml

Purpose	Checkout — address confirmation, delivery selection, voucher input, order summary.
Dependencies	CheckoutController.cs, CheckoutViewModel, _OrderSummary.cshtml, _DeliverySelection.cshtml, checkout.css, checkout.js. Flowcharts: 3_CART_CHECKOUT.mmd, 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	Voucher field: AJAX to VoucherController.ValidateCode on blur. Delivery selection updates shipping fee. Order summary updates live.

Payment.cshtml

Purpose	Payment method selection — GCash, BankTransfer, Cash (POS only in admin).
Dependencies	PaymentController.cs, _PaymentMethod.cshtml, _OTPModal.cshtml, payment.css, payment.js. Flowchart: 4_PAYMENT_PROCESSING.mmd.
Dev Notes	GCash: show QR + reference field. BankTransfer: show account details + file upload for proof. NO card option.

Confirmation.cshtml

Purpose	Order confirmation — order number, itemised summary, estimated delivery.
Dependencies	OrderController.cs, OrderViewModel, _ConfirmationModal.cshtml, confirmation.css, confirmation.js. Flowchart: 5_ORDER_CONFIRMATION_TRACKING.mmd.
Dev Notes	Show confetti animation on load. Display OrderNumber prominently. CTA to OrderHistory.

OrderHistory.cshtml

Purpose	Order history list — status badges, date, total, track button.
Dependencies	OrderController.cs, OrderViewModel, order-history.css, order-history.js. Flowchart: 5_ORDER_CONFIRMATION_TRACKING.mmd.
Dev Notes	Filter by status client-side via order-history.js. Expand/collapse order detail. TrackOrder shows live Delivery status.

Wishlist.cshtml

Purpose	Wishlist page — saved product grid with remove and add-to-cart actions.
Dependencies	WishlistController.cs, WishlistViewModel, wishlist.css, wishlist.js. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	InStock products show add-to-cart. Out-of-stock greyed. Remove is AJAX.

Reviews.cshtml

Purpose	Review submission form and review list for a product.
Dependencies	ReviewController.cs, ReviewViewModel, review.css, review.js. Flowchart: 6_DELIVERY_REVIEW.mmd.
Dev Notes	Only show submit form if IsVerifiedPurchase. Star rating widget driven by review.js. Existing reviews paginated.

Error.cshtml

Purpose	Unhandled exception fallback view — displays RequestId.
Dependencies	HomeController.cs, ErrorViewModel.
Dev Notes	Standard ASP.NET Core error view. Do not add dynamic DB queries here.

_Footer.cshtml

Purpose	Site footer partial — category links, contact, copyright.
Dependencies	_Layout.cshtml, footer.css.
Dev Notes	Static content. Included at bottom of _Layout.cshtml via @await Html.PartialAsync.

_Layout.cshtml

Purpose	Master layout — HTML shell, Bootstrap link from lib/, @RenderBody(), script bundles.
Dependencies	_Navbar.cshtml, _Footer.cshtml, site.css, site.js, wwwroot/lib/bootstrap/.
Dev Notes	Reference Bootstrap from ~/lib/bootstrap/dist/css/bootstrap.min.css (not wwwroot/css/). Include _ValidationScriptsPartial in forms.

_Navbar.cshtml

Purpose	Top navigation — logo, nav links, cart badge, login/logout state.

Dependencies	_Layout.cshtml, CartController (cart count), CustomerController (auth state).
Dev Notes	Cart badge count loaded via AJAX from CartController.GetCartCount. Show/hide auth links based on session.

_ValidationScriptsPartial.cshtml

Purpose	jQuery Validation script bundle for client-side form validation.
Dependencies	Login.cshtml, Register.cshtml, Checkout.cshtml, wwwroot/lib/jquery-validation/.
Dev Notes	Include inside @section Scripts on any form view. Required for data-val-* attribute validation.

_CartItem.cshtml

Purpose	Cart item row partial — thumbnail, name, qty stepper, price, remove.
Dependencies	Cart.cshtml, CartViewModel, cart-item.css. Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	Rendered in a loop. Qty stepper inputs drive cart.js AJAX. Include data-cartitemid attribute.

_ConfirmationModal.cshtml

Purpose	Order success Bootstrap modal — order number, CTA to view order.
Dependencies	Confirmation.cshtml, confirmation-modal.css.
Dev Notes	Auto-triggered on Confirmation.cshtml load via confirmation.js. Shows order number.

_DeliverySelection.cshtml

Purpose	Delivery method radio group — Store Pickup vs Lalamove vs LBC with estimated cost.
Dependencies	Checkout.cshtml, checkout.js, delivery-selection.css. Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	Selection updates ShippingFee in CheckoutViewModel via checkout.js. Store Pickup sets fee=0.

_NotificationAlert.cshtml

Purpose	Toast/banner component — renders TempData messages as Bootstrap alerts.
Dependencies	All views via TempData, notification-alert.css.
Dev Notes	Check TempData[success] and TempData[error] keys. Render dismissible Bootstrap alert.

_OrderSummary.cshtml

Purpose	Order totals panel — line items, subtotal, discount, shipping, grand total.
Dependencies	Checkout.cshtml, Confirmation.cshtml, order-summary.css. Flowcharts: 3_CART_CHECKOUT.mmd, 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	Reusable partial. Grand total updates when voucher applied or delivery method changed.

_OTPModal.cshtml

Purpose	OTP entry modal — code input, resend link, verify button, countdown.
Dependencies	Register.cshtml, Payment.cshtml, otp-modal.css. Flowchart: 1_CUSTOMER_REGISTRATION_LOGIN.mmd.
Dev Notes	Countdown timer in JS. Resend triggers CustomerController.ResendOTP. Verify submits to CustomerController.VerifyOTP.

_PaymentMethod.cshtml

Purpose	Payment method radio cards — GCash, BankTransfer conditional fields.
Dependencies	Payment.cshtml, PaymentController.cs, payment-method.css. Flowchart: 4_PAYMENT_PROCESSING.mmd.
Dev Notes	GCash shows QR + reference number input. BankTransfer shows account details + file upload. NO Card option.

_ProductCard.cshtml

Purpose	Product grid card — thumbnail, name, brand, price, add-to-cart, wishlist toggle.
Dependencies	ProductCatalog.cshtml, ProductViewModel, product-card.css. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Wishlist toggle button is AJAX — calls WishlistController. Stock badge shown when low stock.

site.css

Purpose	Global styles — typography, colour variables, utilities, link styles.
Dependencies	_Layout.cshtml.
Dev Notes	Base layer. Define CSS custom properties (--primary-color etc.) here.

responsive.css

Purpose	Breakpoints for mobile, tablet, desktop.
Dependencies	_Layout.cshtml, site.css.
Dev Notes	Mobile-first. Breakpoints: 576px, 768px, 992px.

auth.css

Purpose	Login and Register page styles.
Dependencies	Login.cshtml, Register.cshtml.
Dev Notes	Form card, input groups, submit button, error text.

customer.css

Purpose	Shared customer area styles — sidebar, breadcrumbs, page headers.
Dependencies	All customer views, _Layout.cshtml.
Dev Notes	Applied across all authenticated customer pages.

dashboard.css

Purpose	Customer dashboard layout — order cards, stats strip, quick actions.
Dependencies	Dashboard.cshtml.
Dev Notes	

cart.css

Purpose	Cart page layout — item list container, totals sidebar, checkout button.
Dependencies	Cart.cshtml.
Dev Notes	

cart-item.css

Purpose	Cart item row — thumbnail, product info, qty stepper, price, remove.
Dependencies	_CartItem.cshtml.
Dev Notes	

checkout.css

Purpose	Checkout page — multi-step form layout, address fields, delivery panel.
Dependencies	Checkout.cshtml.
Dev Notes	

WEBAPPLICATION — CSS (wwwroot/css/)

confirmation.css

Purpose	Confirmation page — success banner, order detail card, next steps.
Dependencies	Confirmation.cshtml.
Dev Notes	

confirmation-modal.css

Purpose	Confirmation modal overlay — backdrop, animation, CTA button.
Dependencies	_ConfirmationModal.cshtml.
Dev Notes	

delivery-selection.css

Purpose	Delivery method radio cards — option highlight, price display.
Dependencies	_DeliverySelection.cshtml.
Dev Notes	

footer.css

Purpose	Footer styles.
Dependencies	_Footer.cshtml.
Dev Notes	Served from wwwroot/css/. Reference via ~/css/footer.css.

Index.css

Purpose	Landing page — hero banner, featured strip, category grid.
Dependencies	Index.cshtml.
Dev Notes	

notification-alert.css

Purpose	Toast/alert banner — success, error, info variants.
Dependencies	_NotificationAlert.cshtml.
Dev Notes	

order-history.css

Purpose	Order history list — card, status badge colours, date/total.
Dependencies	OrderHistory.cshtml.
Dev Notes	

order-summary.css

Purpose	Order totals panel — line items, discount, shipping, grand total.
Dependencies	_OrderSummary.cshtml.
Dev Notes	

otp-modal.css

Purpose	OTP modal — code input boxes, countdown timer, resend link.
Dependencies	OTPModal.cshtml.
Dev Notes	

payment.css

Purpose	Payment page — method panel, QR placeholder, file upload, submit.
Dependencies	Payment.cshtml.
Dev Notes	

payment-method.css

Purpose	Payment method radio cards — icons, selected highlight, conditional fields.
Dependencies	_PaymentMethod.cshtml.
Dev Notes	

product-card.css

Purpose	Product grid card — image hover, title truncation, price, buttons.
Dependencies	_ProductCard.cshtml.
Dev Notes	

product-catalog.css

Purpose	Catalogue page — filter sidebar, grid, sort, pagination.
Dependencies	ProductCatalog.cshtml.
Dev Notes	

product-details.css

Purpose	Product detail — image gallery, spec table, variant selector.
Dependencies	ProductDetails.cshtml.
Dev Notes	

profile.css

Purpose	Profile page — info card, edit form, address section, change password.
Dependencies	Profile.cshtml.
Dev Notes	

register.css

Purpose	Registration form — multi-field card, validation highlights.
Dependencies	Register.cshtml.
Dev Notes	

review.css

Purpose	Review component — star rating widget, review cards, submission form.
Dependencies	Reviews.cshtml, ProductDetails.cshtml.
Dev Notes	

wishlist.css

Purpose	Wishlist page — saved product grid, remove button, add-to-cart action.
Dependencies	Wishlist.cshtml.
Dev Notes	

site.js

Purpose	Global JS init — Bootstrap tooltips/popovers, navbar scroll, global event listeners.
Dependencies	_Layout.cshtml.
Dev Notes	Runs on every page. Keep lightweight.

utils.js

Purpose	Shared utilities — formatCurrency, debounce, throttle, showAlert, fetchWithCSRF.
Dependencies	All other JS files.
Dev Notes	Import via @section Scripts. fetchWithCSRF wraps fetch with the CSRF token header.

customer.js

Purpose	Customer area JS — profile dropdown, notification bell, session expiry warning.
Dependencies	_Navbar.cshtml, customer.css.
Dev Notes	Loaded on all authenticated customer pages.

cart.js

Purpose	Cart JS — qty stepper handlers, AJAX remove, real-time subtotal, cart badge refresh.
Dependencies	Cart.cshtml, CartController.cs (AJAX). Flowchart: 3_CART_CHECKOUT.mmd.
Dev Notes	POST to CartController.UpdateQuantity and RemoveFromCart. Update #cart-badge count in navbar.

checkout.js

Purpose	Checkout JS — address validation, voucher AJAX, delivery price update.
Dependencies	Checkout.cshtml, VoucherController.cs, checkout.css. Flowcharts: 3_CART_CHECKOUT.mmd, 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	On voucher code blur: POST to VoucherController.ValidateCode, update discount line in order summary. Delivery method change: update shipping fee display.

confirmation.js

Purpose	Confirmation JS — success animation, print receipt, confetti.
Dependencies	Confirmation.cshtml.
Dev Notes	Trigger confetti on DOMContentLoaded. Print via window.print().

order-history.js

Purpose	Order history JS — client-side filter by status, date range, expand/collapse.
Dependencies	OrderHistory.cshtml, OrderController.cs. Flowchart: 5_ORDER_CONFIRMATION_TRACKING.mmd.
Dev Notes	Filter without full page reload. Expand row to show order items inline.

payment.js

Purpose	Payment JS — method toggle, OTP send/verify AJAX, GCash redirect.

Dependencies	Payment.cshtml, PaymentController.cs. Flowchart: 4_PAYMENT_PROCESSING.mmd.
Dev Notes	Method toggle: show/hide GCash vs BankTransfer conditional sections. OTP AJAX to CustomerController.SendOTP.

WEBAPPLICATION — JavaScript (wwwroot/js/)

product-catalog.js

Purpose	Catalogue JS — filter AJAX reload, search debounce, sort, wishlist toggle.
Dependencies	ProductCatalog.cshtml, ProductController.cs, WishlistController.cs. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Debounce search input (300ms). Filter and sort POST to ProductController.List. Wishlist toggle updates heart icon and count.

review.js

Purpose	Review JS — star-rating widget interaction, AJAX review submit, pagination.
Dependencies	Reviews.cshtml, ReviewController.cs. Flowchart: 6_DELIVERY_REVIEW.mmd.
Dev Notes	Click-to-set star rating sets hidden Rating input. Submit via AJAX, append new review card without reload.

wishlist.js

Purpose	Wishlist JS — AJAX remove, add-to-cart from wishlist page.
Dependencies	Wishlist.cshtml, WishlistController.cs, CartController.cs. Flowchart: 2_CUSTOMER_DASHBOARD_SHOPPING.mmd.
Dev Notes	Remove calls WishlistController.Remove AJAX. Add-to-cart calls CartController.AddToCart AJAX.

ADMINSYSTEM — Configuration

AdminSystem.csproj

Purpose	MSBuild project file — .NET Framework 4.8 WPF application definition.
Dependencies	App.config, App.xaml, all .cs and .xaml files.
Dev Notes	Target: .NET Framework 4.8. References: Dapper, BCrypt.Net, WPF assemblies.

App.config

Purpose	Application configuration — SQL Server connection string for direct ADO.NET/Dapper access.
Dependencies	DatabaseHelper.cs, all Repository classes.
Dev Notes	Connection string points to TaurusBikeShopDB. This is the only config file for AdminSystem.

App.xaml

Purpose	WPF application entry point — declares application-level resources and startup window.
Dependencies	App.xaml.cs, AppStyles.xaml.
Dev Notes	Merged resource dictionary: include AppStyles.xaml here for global styles.

App.xaml.cs

Purpose	Code-behind for App.xaml — bootstraps DI container and sets startup window.
Dependencies	App.xaml, Services/, ViewModels/, LoginViewModel.
Dev Notes	Register all services and ViewModels. Set StartupUri to Login.xaml.

AppStyles.xaml

Purpose	Shared WPF ResourceDictionary — button styles, colours, fonts for consistent admin UI.
Dependencies	App.xaml, all Views.
Dev Notes	Define named brushes (PrimaryBrush, AccentBrush) here. Reference in Views via StaticResource.

ADMINSYSTEM — Converters & Helpers

BoolToVisibility.cs

Purpose	IValueConverter — bool to Visibility (Visible/Collapsed) for XAML bindings.
Dependencies	All Views that toggle element visibility based on ViewModel bool properties.
Dev Notes	Register in AppStyles.xaml as a StaticResource for reuse across all views.

CurrencyConverter.cs

Purpose	IValueConverter — decimal price to formatted PHP string (₱1,234.00).
Dependencies	POS.xaml, Inventory.xaml, Orders.xaml, Reports.xaml.
Dev Notes	Use Philippine Peso symbol. Format: ₱#,##0.00.

StatusToColor.cs

Purpose	IValueConverter — order/delivery status string to SolidColorBrush for status badges.
Dependencies	Orders.xaml, Delivery.xaml.
Dev Notes	Map: Pending=amber, Processing=blue, Delivered=green, Cancelled=red, OnHold=orange.

DatabaseHelper.cs

Purpose	ADO.NET/Dapper SQL Server connection helper — reads connection string from App.config.
Dependencies	App.config, all Repository classes in Repositories/.
Dev Notes	Provides GetConnection() returning open SqlConnection. All queries go through this.

NavigationHelper.cs

Purpose	WPF navigation — manages page transitions within MainWindow content frame.
Dependencies	MainWindow.xaml.cs, all Views.
Dev Notes	NavigateTo<T>(): load View into MainWindow content frame. Pass ViewModel as DataContext.

PasswordHelper.cs

Purpose	BCrypt hash verification for admin login authentication.
Dependencies	AuthService.cs, LoginViewModel.cs.
Dev Notes	Verify(input, hash): BCrypt.Net.BCrypt.Verify. Mirrors WebApplication PasswordHelper.

ADMINSYSTEM — Models (18 POCO Classes)

PrintHelper.cs

Purpose	POS receipt printing using WPF PrintDialog.
Dependencies	POS.xaml.cs, POSViewModel.cs.
Dev Notes	GenerateReceipt(order): formats receipt as FlowDocument, sends to PrintDialog.

AdminSystem/Models/BankTransferPayment.cs

Purpose	POCO mirroring BankTransferPayment table for admin payment verification.
Dependencies	PaymentRepository.cs, PaymentService.cs, Orders.xaml. Flowcharts: 9_PAYMENT_VERIFICATION.mmd.
Dev Notes	Includes VerificationDeadline for auto-hold display and VerificationNotes for rejection reason.

AdminSystem/Models/Brand.cs

Purpose	POCO mirroring Brand table.
Dependencies	ProductRepository.cs, ProductViewModel.cs, Inventory.xaml.
Dev Notes	

AdminSystem/Models/Category.cs

Purpose	POCO mirroring Category table.
Dependencies	ProductRepository.cs, POSViewModel.cs, InventoryViewModel.cs.
Dev Notes	

AdminSystem/Models/Delivery.cs

Purpose	POCO mirroring Delivery table with courier fields.
Dependencies	DeliveryService.cs, DeliveryViewModel.cs, Delivery.xaml. Flowcharts: 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	IsDelayed + DelayedUntil shown in logistics board. Include LalamoveDelivery and LBCDelivery sub-objects.

AdminSystem/Models/GCashPayment.cs

Purpose	POCO mirroring GCashPayment subtype table.

Dependencies	PaymentRepository.cs, PaymentService.cs. Flowchart: 9_PAYMENT_VERIFICATION.mmd.
Dev Notes	GcashTransactionId displayed in payment verification screen.

AdminSystem/Models/InventoryLog.cs

Purpose	POCO mirroring InventoryLog table.
Dependencies	InventoryService.cs, InventoryViewModel.cs, Inventory.xaml. Flowchart: 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	ChangeType display label in Inventory audit log view.

AdminSystem/Models/LalamoveDelivery.cs

Purpose	POCO mirroring LalamoveDelivery table.
Dependencies	DeliveryService.cs, DeliveryViewModel.cs. Flowchart: 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	BookingRef, DriverName, DriverPhone displayed in logistics board when available.

AdminSystem/Models/LBCDelivery.cs

Purpose	POCO mirroring LBCDelivery table.
Dependencies	DeliveryService.cs, DeliveryViewModel.cs. Flowchart: 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	TrackingNumber for LBC waybill display.

AdminSystem/Models/Order.cs

Purpose	POCO mirroring Order table for admin order management.
Dependencies	OrderRepository.cs, OrderService.cs, OrderViewModel.cs, Orders.xaml. Flowcharts: 8_ONLINE_ORDERS_MANAGEMENT.mmd.
Dev Notes	Include IsWalkIn to differentiate POS and online orders. TotalAmount computed client-side.

AdminSystem/Models/OrderItem.cs

Purpose	POCO mirroring OrderItem table.
Dependencies	OrderRepository.cs, POSViewModel.cs, POS.xaml.
Dev Notes	UnitPrice snapshot. Include ProductName for display.

AdminSystem/Models/Payment.cs

Purpose	POCO mirroring Payment table for admin side.
Dependencies	PaymentRepository.cs, PaymentService.cs, POSViewModel.cs. Flowcharts: 9_PAYMENT_VERIFICATION.mmd, 11_WALKIN_POS.mmd.
Dev Notes	PaymentMethod: GCash, BankTransfer, Cash only — NO Card.

AdminSystem/Models/PickupOrder.cs

Purpose	POCO mirroring PickupOrder table.
Dependencies	OrderService.cs, OrderViewModel.cs, Orders.xaml. Flowchart: 8_ONLINE_ORDERS_MANAGEMENT.mmd.
Dev Notes	PickupExpiresAt displayed in pickup orders list. Set PickupReadyAt when admin marks ReadyForPickup.

AdminSystem/Models/POS_Session.cs

Purpose	POCO mirroring POS_Session table.
Dependencies	POSViewModel.cs, POS.xaml. Flowchart: 11_WALKIN_POS.mmd.
Dev Notes	ShiftEnd NULL while active. TotalSales running total per session.

AdminSystem/Models/Product.cs

Purpose	POCO mirroring Product table for POS and inventory.
Dependencies	ProductRepository.cs, ProductService.cs, POSViewModel.cs, InventoryViewModel.cs. Flowcharts: 11_WALKIN_POS.mmd, 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	No StockQuantity — use ProductVariant for stock. IsFeatured for homepage management.

AdminSystem/Models/Review.cs

Purpose	POCO mirroring Review table for admin review moderation.
Dependencies	OrderRepository.cs, Users.xaml.
Dev Notes	Rating 1-5. IsVerifiedPurchase badge in moderation screen.

AdminSystem/Models/SupportTask.cs

Purpose	POCO mirroring SupportTask table (v7.1).
Dependencies	SupportService.cs, SupportViewModel.cs, Support.xaml. Flowchart: 13_SUPPORT_SYSTEM.mmd.
Dev Notes	TaskType and TaskStatus CHECK values enforced. CompletedAt set when Done.

AdminSystem/Models/User.cs

Purpose	POCO mirroring User table for admin auth and user management.
Dependencies	AuthService.cs, LoginViewModel.cs, UserRepository.cs, Users.xaml. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	IsWalkIn=true for POS placeholder users. Role loaded separately via UserRole.

AdminSystem/Models/Voucher.cs

Purpose	POCO mirroring Voucher table for POS discount application.
Dependencies	POSViewModel.cs, POS.xaml, Vouchers.xaml. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	POS staff apply vouchers by code. Same validation rules as web: date range, MaxUses, etc.

ADMINSYSTEM — Repositories

AdminSystem/Repositories/IRepository.cs

Purpose	Generic repository interface for AdminSystem.
Dependencies	Repository.cs, all AdminSystem-specific repositories.
Dev Notes	Mirrors WebApplication IRepository pattern but uses ADO.NET/Dapper instead of EF Core.

AdminSystem/Repositories/Repository.cs

Purpose	Generic Dapper repository base for AdminSystem.
Dependencies	IRepository.cs, DatabaseHelper.cs.
Dev Notes	GetAll, GetById, Add, Update, Delete via raw SQL with Dapper mapping.

AdminSystem/Repositories/InventoryRepository.cs

Purpose	Inventory data access — InventoryLog reads, ProductVariant stock updates.
Dependencies	Repository.cs, InventoryLog.cs, Product.cs, DatabaseHelper.cs. Flowcharts: 12_INVENTORY_MANAGEMENT.mmd, 16_SUPPLIER_PURCHASE_ORDER.mmd.
Dev Notes	AdjustStock: UPDATE ProductVariant StockQuantity, INSERT InventoryLog. Used by both InventoryService and SupplierService.

AdminSystem/Repositories/OrderRepository.cs

Purpose	Order and OrderItem CRUD for admin order management.
Dependencies	Repository.cs, Order.cs, OrderItem.cs, PickupOrder.cs, DatabaseHelper.cs. Flowcharts: 8_ONLINE_ORDERS_MANAGEMENT.mmd.
Dev Notes	GetPendingOrders, UpdateOrderStatus, SetPickupReady (sets PickupReadyAt + PickupExpiresAt).

AdminSystem/Repositories/PaymentRepository.cs

Purpose	Payment data access for admin verification and POS recording.
Dependencies	Repository.cs, Payment.cs, BankTransferPayment.cs, GCashPayment.cs, DatabaseHelper.cs. Flowcharts: 9_PAYMENT_VERIFICATION.mmd, 11_WALKIN_POS.mmd.
Dev Notes	GetPendingVerifications: JOIN Payment + BankTransferPayment WHERE Status=VerificationPending. Verify/Reject: update VerifiedAt, VerifiedByUserId, VerificationNotes.

AdminSystem/Repositories/ProductRepository.cs

Purpose	Product queries for POS barcode scan and inventory management.
Dependencies	Repository.cs, Product.cs, Brand.cs, Category.cs, DatabaseHelper.cs. Flowcharts: 11_WALKIN_POS.mmd, 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	GetBySKU: fast barcode lookup. GetWithVariants: returns Product + all ProductVariant rows.

AdminSystem/Repositories/UserRepository.cs

Purpose	User lookup and role validation for admin authentication.
Dependencies	Repository.cs, User.cs, DatabaseHelper.cs. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	FindByEmail + role join. Only Admin, Manager, Cashier, Staff roles can access AdminSystem.

ADMINSYSTEM — Services & Interfaces

AdminSystem/Services/IAuthService.cs

Purpose	Interface for admin authentication — Login, ValidateSession.
Dependencies	AuthService.cs, LoginViewModel.cs. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	Returns authenticated User on success. ValidateSession checks session token validity.

AdminSystem/Services/IDeliveryService.cs

Purpose	Interface for logistics — GetPendingDeliveries, DispatchOrder, MarkDelayed, CancelDelivery.
Dependencies	DeliveryService.cs, DeliveryViewModel.cs. Flowcharts: 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	DispatchOrder: call Lalamove or LBC API, store booking ref, update status to Shipped.

AdminSystem/Services/IInventoryService.cs

Purpose	Interface for stock management — AdjustStock, GetLowStockItems, GetInventoryLog.
Dependencies	InventoryService.cs, InventoryViewModel.cs. Flowchart: 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	AdjustStock: writes InventoryLog and updates ProductVariant.StockQuantity atomically.

AdminSystem/Services/INotificationService.cs

Purpose	Interface for queueing notifications from AdminSystem.
Dependencies	NotificationService.cs, SupportService.cs. Flowcharts: 12_INVENTORY_MANAGEMENT.mmd, 13_SUPPORT_SYSTEM.mmd.
Dev Notes	Same queue pattern as WebApplication — inserts Notification row in shared DB.

AdminSystem/Services/IOrderService.cs

Purpose	Interface for admin order management — GetOrders, UpdateStatus, CreateWalkInOrder.
Dependencies	OrderService.cs, OrderViewModel.cs, POSViewModel.cs. Flowcharts: 8_ONLINE_ORDERS_MANAGEMENT.mmd, 11_WALKIN_POS.mmd.
Dev Notes	CreateWalkInOrder: writes Order (IsWalkIn=true), OrderItems, Payment in a transaction.

AdminSystem/Services/IPaymentService.cs

Purpose	Interface for admin payment — ProcessCash, ProcessGCash, RecordPayment. NO ProcessCard.
Dependencies	PaymentService.cs, POSViewModel.cs. Flowcharts: 9_PAYMENT_VERIFICATION.mmd, 11_WALKIN_POS.mmd.
Dev Notes	Cash and GCash only for POS. BankTransfer verification is admin-only. ProcessCard must NOT exist.

AdminSystem/Services/IProductService.cs

Purpose	Interface for product operations in admin context — search, CRUD, stock level reads.
Dependencies	ProductService.cs, POSViewModel.cs, InventoryViewModel.cs. Flowcharts: 11_WALKIN_POS.mmd, 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	SearchBySKU for barcode scan. UpdateProduct for catalogue maintenance.

AdminSystem/Services/IReportService.cs

Purpose	Interface for reporting — GetSalesSummary, GetInventoryReport, GetRevenueByDate.
Dependencies	ReportService.cs, ReportViewModel.cs. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	Returns aggregated data for dashboard KPIs and report charts.

AdminSystem/Services/ISupportService.cs

Purpose	Interface for admin support ticket management — view, reply, create task, resolve.
Dependencies	SupportService.cs, SupportViewModel.cs. Flowchart: 13_SUPPORT_SYSTEM.mmd.
Dev Notes	ReplyToTicket: insert SupportTicketReply, update TicketStatus to AwaitingResponse. CreateTask: insert SupportTask.

AdminSystem/Services/AuthService.cs

Purpose	Implements IAuthService — validates admin credentials via User table using PasswordHelper.
Dependencies	IAuthService.cs, UserRepository.cs, PasswordHelper.cs, DatabaseHelper.cs.
Dev Notes	Only allow users with Admin, Manager, Cashier, or Staff roles. Return User with role info.

AdminSystem/Services/DeliveryService.cs

Purpose	Implements IDeliveryService — queries Delivery, dispatches via courier APIs, updates status.
Dependencies	IDeliveryService.cs, DeliveryViewModel.cs, Delivery.cs, DatabaseHelper.cs. Flowchart: 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	MarkDelayed: set IsDelayed=true and DelayedUntil, trigger delay notification via INotificationService.

AdminSystem/Services/InventoryService.cs

Purpose	Implements IInventoryService — stock adjustments and InventoryLog writes.
Dependencies	IInventoryService.cs, InventoryRepository.cs, InventoryLog.cs, DatabaseHelper.cs. Flowchart: 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	Every stock change goes through this service. Check ReorderThreshold after each adjustment.

AdminSystem/Services/NotificationService.cs

Purpose	Implements INotificationService for AdminSystem — writes Notification rows to shared DB.
Dependencies	INotificationService.cs, DatabaseHelper.cs.
Dev Notes	Uses same Notification table as WebApplication. Inserts row; dispatch is handled by background job in WebApplication.

AdminSystem/Services/OrderService.cs

Purpose	Implements IOrderService — creates walk-in orders, updates online statuses.
Dependencies	IOrderService.cs, OrderRepository.cs, Order.cs, PaymentService.cs. Flowcharts: 8_ONLINE_ORDERS_MANAGEMENT.mmd, 11_WALKIN_POS.mmd.
Dev Notes	CreateWalkInOrder: IsWalkIn=true, ShippingAddressId=null, instant status=Delivered. Write InventoryLog Sale.

AdminSystem/Services/PaymentService.cs

Purpose	Implements IPaymentService — records Cash and GCash POS payments.
Dependencies	IPaymentService.cs, PaymentRepository.cs, Payment.cs. Flowchart: 11_WALKIN_POS.mmd.
Dev Notes	POS cash: PaymentMethod=Cash, PaymentStatus=Completed immediately. GCash: call external gateway, store GcashTransactionId.

AdminSystem/Services/ProductService.cs

Purpose	Implements IProductService for admin — product CRUD and barcode lookup.
Dependencies	IProductService.cs, ProductRepository.cs, Product.cs. Flowcharts: 11_WALKIN_POS.mmd, 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	SearchBySKU used by POS barcode scan for instant item lookup.

AdminSystem/Services/ReportService.cs

Purpose	Implements IReportService — aggregated SQL queries for KPIs and charts.
Dependencies	IReportService.cs, DatabaseHelper.cs, multiple Models. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	Queries span Order, Payment, InventoryLog, PurchaseOrder tables. Returns DTOs for chart binding.

AdminSystem/Services/SupportService.cs

Purpose	Implements ISupportService — ticket view, reply, task creation, resolve.
Dependencies	ISupportService.cs, DatabaseHelper.cs, SupportTicket.cs, SupportTask.cs, INotificationService.cs. Flowchart: 13_SUPPORT_SYSTEM.mmd.
Dev Notes	ReplyToTicket: INSERT SupportTicketReply, UPDATE TicketStatus. CreateTask: INSERT SupportTask. Resolve: SET ResolvedAt=now, TicketStatus=Resolved.

AdminSystem/ViewModels/BaseViewModel.cs

Purpose	Abstract MVVM base — INotifyPropertyChanged, SetProperty<T> helper.
Dependencies	All other AdminSystem ViewModels.
Dev Notes	All VMs inherit this. SetProperty: compare value, set backing field, raise PropertyChanged.

AdminSystem/ViewModels/DashboardViewModel.cs

Purpose	Admin dashboard — KPI counters, recent activity, navigation commands.
Dependencies	BaseViewModel.cs, ReportService.cs, Dashboard.xaml. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	Load: TodaySales, PendingOrders, LowStockCount, OpenTickets. Bind to Dashboard.xaml metric tiles.

ADMINSYSTEM — ViewModels (12 VMs)

AdminSystem/ViewModels/DeliveryViewModel.cs

Purpose	Delivery board — pending dispatches, dispatch command, delay/cancel commands.
Dependencies	BaseViewModel.cs, DeliveryService.cs, Delivery.cs, Delivery.xaml. Flowchart: 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	DispatchCommand: call DeliveryService.DispatchOrder, refresh list. MarkDelayedCommand: prompt for new ETA.

AdminSystem/ViewModels/InventoryViewModel.cs

Purpose	Stock management — product list, adjustment form, low-stock filter, audit log tab.
Dependencies	BaseViewModel.cs, InventoryService.cs, InventoryLog.cs, Inventory.xaml. Flowcharts: 12_INVENTORY_MANAGEMENT.mmd, 16_SUPPLIER_PURCHASE_ORDER.mmd.
Dev Notes	AdjustStockCommand: validate qty, call InventoryService.AdjustStock, refresh grid. Tab switch for audit log vs stock levels.

AdminSystem/ViewModels/LoginViewModel.cs

Purpose	Binds username/password — LoginCommand calls AuthService.Login.
Dependencies	BaseViewModel.cs, AuthService.cs, PasswordHelper.cs, Login.xaml. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	LoginCommand: validate fields, call AuthService, navigate to MainWindow on success, show error on fail.

AdminSystem/ViewModels/OrderViewModel.cs

Purpose	Order list with filters, status update command, detail panel.
Dependencies	BaseViewModel.cs, OrderService.cs, Order.cs, PickupOrder.cs, Orders.xaml. Flowcharts: 8_ONLINE_ORDERS_MANAGEMENT.mmd.
Dev Notes	StatusUpdateCommand: call OrderService.UpdateStatus, refresh list. SetPickupReadyCommand: set PickupReadyAt.

AdminSystem/ViewModels/POSViewModel.cs

Purpose	POS session — item search, cart list, totals, payment modal, receipt.
Dependencies	BaseViewModel.cs, OrderService.cs, PaymentService.cs, ProductService.cs, POS.xaml, PrintHelper.cs. Flowchart: 11_WALKIN_POS.mmd.
Dev Notes	Most complex VM. SearchCommand: barcode/SKU lookup. AddToCartCommand. PaymentMethod: Cash or GCash only. CompleteCommand: create walk-in order + payment + print receipt.

AdminSystem/ViewModels/ProductViewModel.cs

Purpose	Product CRUD — add/edit product, variant management, brand/category bind.
Dependencies	BaseViewModel.cs, ProductService.cs, Product.cs, Brand.cs, Category.cs, Products.xaml. Flowchart: 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	SaveCommand: validate form, call ProductService.CreateProduct or UpdateProduct.

AdminSystem/ViewModels/ReportViewModel.cs

Purpose	Report type selection, date range, chart data, export command.
Dependencies	BaseViewModel.cs, ReportService.cs, Reports.xaml. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	GenerateReportCommand: call ReportService with date range, bind DataGrid and chart. ExportCommand: CSV or print.

AdminSystem/ViewModels/SupportViewModel.cs

Purpose	Support ticket management — list, filter by status, reply, task, resolve.
Dependencies	BaseViewModel.cs, SupportService.cs, SupportTask.cs, Support.xaml. Flowchart: 13_SUPPORT_SYSTEM.mmd.

Dev Notes	ReplyCommand: open reply input, call SupportService.ReplyToTicket. CreateTaskCommand: show task form dialog. ResolveCommand: confirm, call SupportService.ResolveTicket.

AdminSystem/ViewModels/UsersViewModel.cs

Purpose	User and staff account management — list, role assignment, enable/disable.
Dependencies	BaseViewModel.cs, UserRepository.cs, User.cs, Users.xaml. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	AssignRoleCommand: insert/delete UserRole. ToggleActiveCommand: UPDATE User.IsActive.

AdminSystem/ViewModels/VoucherViewModel.cs

Purpose	Voucher management — list, create, activate/deactivate.
Dependencies	BaseViewModel.cs, Voucher.cs, Vouchers.xaml. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	CreateCommand: validate Code uniqueness, DiscountType, date range. ToggleActiveCommand: flip IsActive.

Login.xaml / .cs

Purpose	Admin login screen — username, password, login button, error label.
Dependencies	LoginViewModel.cs, AuthService.cs. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	Startup view. Successful login navigates to MainWindow. Show error on invalid credentials.

MainWindow.xaml / .cs

Purpose	Application shell — sidebar navigation menu, content frame, status bar.
Dependencies	NavigationHelper.cs, DashboardViewModel.cs, all other Views. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	Content frame loads Views on menu click via NavigationHelper.NavigateTo<T>(). Status bar shows logged-in user name and role.

ADMINSYSTEM — Views (12 XAML Pairs)

Dashboard.xaml / .cs

Purpose	Admin dashboard — KPI tiles (orders, revenue, low-stock), recent activity.
Dependencies	DashboardViewModel.cs, ReportService.cs. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	First screen after login. Bind metric tiles to DashboardViewModel properties.

POS.xaml / .cs

Purpose	Walk-in POS — barcode/SKU search, item DataGrid, totals, payment modal.
Dependencies	POSViewModel.cs, PrintHelper.cs, PaymentService.cs. Flowchart: 11_WALKIN_POS.mmd.
Dev Notes	Most complex view. Search box triggers ProductService.SearchBySKU. Payment modal toggles Cash/GCash. Receipt generation on completion.

Inventory.xaml / .cs

Purpose	Inventory management — stock DataGrid, adjustment form, audit log tab.
Dependencies	InventoryViewModel.cs, InventoryService.cs. Flowcharts: 12_INVENTORY_MANAGEMENT.mmd, 16_SUPPLIER_PURCHASE_ORDER.mmd.
Dev Notes	Tab 1: stock levels with ReorderThreshold highlight. Tab 2: InventoryLog audit trail.

Orders.xaml / .cs

Purpose	Online order management — order list, status update, pickup and delivery actions.

Dependencies	OrderViewModel.cs, OrderService.cs. Flowcharts: 8_ONLINE_ORDERS_MANAGEMENT.mmd.
Dev Notes	Filter by OrderStatus. PickupOrders section: set ReadyForPickup, confirm collection. Delivery section: dispatch via courier APIs.

Products.xaml / .cs

Purpose	Product catalogue maintenance — product DataGrid, add/edit form, variant management.
Dependencies	ProductViewModel.cs, ProductService.cs. Flowchart: 12_INVENTORY_MANAGEMENT.mmd.
Dev Notes	Inline edit for stock. Add variant button opens sub-form. IsFeatured toggle for homepage.

Reports.xaml / .cs

Purpose	Reports — date range picker, report type selector, chart canvas, export button.
Dependencies	ReportViewModel.cs, ReportService.cs. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	Chart control: use LiveCharts or similar WPF chart library. Export to CSV via StreamWriter.

Support.xaml / .cs

Purpose	Support ticket management — ticket list, detail panel, reply, task creation, resolve.
Dependencies	SupportViewModel.cs, SupportService.cs. Flowchart: 13_SUPPORT_SYSTEM.mmd.
Dev Notes	Left: ticket list with status filter. Right: ticket detail + reply thread + action buttons.

Users.xaml / .cs

Purpose	User/staff account management — list, role assignment, enable/disable toggle.
Dependencies	UsersViewModel.cs, UserRepository.cs. Flowchart: 7_ADMIN_OVERVIEW.mmd.
Dev Notes	Show role badges. Role assignment via CheckBox per role row. IsActive toggle.

Vouchers.xaml / .cs

Purpose	Voucher management — list, create form (code, type, value, dates), activate/deactivate.
Dependencies	VoucherViewModel.cs, Voucher.cs. Flowchart: 15_VOUCHER_CHECKOUT.mmd.
Dev Notes	Validate Code uniqueness on save. DiscountType radio: Percentage vs Fixed. Date range pickers.

Delivery.xaml / .cs

Purpose	Logistics board — pending delivery list, dispatch action, delay flag, status tracker.
Dependencies	DeliveryViewModel.cs, DeliveryService.cs. Flowchart: 10_LOGISTICS_MANAGEMENT.mmd.
Dev Notes	Show LalamoveDelivery.DriverName when assigned. LBC shows TrackingNumber. Mark Delayed opens ETA update dialog.

SECTION 2 — Chronological Implementation Guide

Implement in the order below to avoid dependency conflicts. Files marked as parallel can be worked on simultaneously by different developers or in separate branches.

STEP 1 Database Schema — run Taurus_schema_v7.1.sql	
Reason	All 38 tables, 12 views, 1 trigger must exist before any C# code touches the DB.
Depends on	SQL Server instance running. TaurusBikeShopDB database created.
Parallel?	Run seed file (Taurus_seed_v7.1.sql) in parallel to populate catalog data.

STEP 2 Seed Data — run Taurus_seed_v7.1.sql	
Reason	150 products, 43 brands, 13 categories must exist to test catalog and POS features.
Depends on	Step 1 (schema) complete.
Parallel?	Can run Taurus_seed.sql (demo transactions) in parallel after schema is ready.

STEP 3 WebApplication Config — Program.cs, AppDbContext.cs, appsettings.json	
Reason	DI container, DB connection, and EF Core mappings must be wired before any feature works.
Depends on	Steps 1-2 complete. .NET 8 SDK installed. EF Core NuGet packages added to WebApplication.csproj.
Parallel?	AdminSystem App.config can be configured in parallel.

STEP 4 All 38 Entity Classes (Models/Entities/)	
Reason	Every service, repository, and controller depends on entity classes. Must exist before anything else compiles.
	Step 3 (AppDbContext) ready to register DbSet<T> entries.
Parallel?	All 38 entity files can be written in parallel — they have no inter-dependencies.

STEP 5 AppDbContext — add all 38 DbSet<T> and OnModelCreating 1:1 configs	
Reason	EF Core needs the 5 shared-PK 1:1 relationships configured (GCashPayment, BankTransferPayment, LalamoveDelivery, LBCDelivery, PickupOrder).
Depends on	Step 4 (all entities) complete.
Parallel?	Can begin writing repository interfaces in parallel.

STEP 6 Utilities — PasswordHelper.cs, ValidationHelper.cs, FileUploadHelper.cs	
Reason	PasswordHelper is needed by UserService (registration/login). Others needed by controllers.
Depends on	No dependencies.
Parallel?	All 3 utilities can be written in parallel and independently.

STEP 7 Generic Repository — IRepository.cs, Repository.cs	
Reason	All specific repositories inherit from Repository.cs. Must exist first.
Depends on	Step 5 (AppDbContext) complete.
	IRepository and Repository can be written simultaneously.

STEP 8 Core Repositories — UserRepository, ProductRepository, BrandRepository, CategoryRepository (via ProductRepository)	
Reason	UserService (auth/registration) and ProductService (catalog, POS) are the earliest features needed.
Depends on	Steps 5-7 complete.
	UserRepository and ProductRepository can be written in parallel.

STEP 9 Auth — IUserService, UserService, IPasswordHelper, CustomerController (Register/Login/OTP)	
	Registration and login must work before any authenticated feature can be tested. Flowchart: Part 1.
Depends on	Steps 6-8 complete.
Parallel?	Can write IUserService interface and CustomerController shell in parallel while UserService is implemented.

STEP 10 Product Catalog — IProductService, ProductService, IBrandService, BrandService, ProductController, ProductCatalog.cshtml, ProductDetails.cshtml	
	Shopping flow depends on product browsing. Flowchart: Part 2.
Depends on	Steps 8-9 complete.
Parallel?	ProductController and ProductCatalog.cshtml can be written in parallel with ProductService.

STEP 11 Cart System — ICartService, CartService, CartRepository, CartController, Cart.cshtml	
	Checkout, voucher, and order flow all depend on an active cart. Flowchart: Part 3.
Depends on	Steps 9-10 complete.
Parallel?	CartController and Cart.cshtml can be written in parallel with CartService.

STEP 12 Voucher System — IVoucherService, VoucherService, VoucherRepository, VoucherController	
	Voucher validation is part of checkout. Must exist before CheckoutController. Flowchart: Part 15.
Depends on	Step 11 (cart) complete.
Parallel?	VoucherController AJAX endpoint can be stubbed and wired to checkout.js independently.

STEP 13 Checkout + Order Creation — IOrderService, OrderService, OrderRepository, CheckoutController, Checkout.cshtml, _OrderSummary.cshtml, _DeliverySelection.cshtml	
Reason	Checkout orchestrates cart + voucher + order creation. Flowcharts: Parts 3, 15.
Depends on	Steps 11-12 complete. InventoryLog Lock write needed inside OrderService.CreateOrder.
Parallel?	Checkout.cshtml and checkout.js can be written in parallel with OrderService.

STEP 14 Payment Processing — IPaymentService, PaymentService, PaymentRepository, PaymentController, Payment.cshtml, _PaymentMethod.cshtml	
Reason	Payment must come after order creation. Flowchart: Part 4.
Depends on	Step 13 complete.
Parallel?	Payment.cshtml and payment.js can be written in parallel with PaymentService.

STEP 15 Order Confirmation + Tracking — OrderController, Confirmation.cshtml, OrderHistory.cshtml	
Reason	Post-payment confirmation and customer order tracking. Flowchart: Part 5.
Depends on	Step 14 complete.
Parallel?	Confirmation.cshtml and OrderHistory.cshtml can be written in parallel.

STEP 16 Wishlist — IWishlistService, WishlistService, WishlistRepository, WishlistController, Wishlist.cshtml	
	Wishlist is independent of checkout but needed for Part 2 and Part 12 (stock restore notifications).
Depends on	Steps 9-10 complete (user and product must exist).
Parallel?	Entirely parallel to Steps 12-15.

STEP 17 Reviews — IReviewService, ReviewService, ReviewRepository, ReviewController, Reviews.cshtml, review.js	
	Post-delivery review submission. Flowchart: Part 6.
Depends on	Steps 14-15 complete (order must exist and be Delivered).

Parallel?	ReviewService and ReviewController can be written in parallel.

STEP 18 Support Tickets (Web) — ISupportService, SupportService, SupportRepository, SupportController	
	Customers create tickets from Dashboard (Part 2) and Delivery review (Part 6). Flowchart: Part 13.
Depends on	Steps 9, 15 complete.
Parallel?	Can be written in parallel with Steps 16-17.

STEP 19 Notification Service — INotificationService, NotificationService	
	All services that queue notifications depend on this. Should be implemented early but can be stubbed initially.
Depends on	Step 4 (Notification entity) complete.
Parallel?	Write the Notification entity and NotificationService shell early; fill in dispatch logic later.

STEP 20 Background Jobs — all 5 IHostedService classes, register in Program.cs	
Reason	Automated monitoring requires all entities and services to be complete. Flowchart: Part 14.
Depends on	Steps 13-19 complete. All entities, services, and DB tables in place.
Parallel?	Each of the 5 jobs can be written in parallel with each other.

STEP 21 Supplier & Purchase Order (Web read-only) — SupplierRepository	
Reason	SupplierRepository is needed by AdminSystem stock receiving. Web side reads only. Flowchart: Part 16.
Depends on	Steps 4-7 complete.
	Can be written in parallel with Steps 16-18.

STEP 22 AdminSystem — Config, DatabaseHelper, PasswordHelper, Repositories	
Reason	Foundation layer for AdminSystem. All admin services depend on DatabaseHelper and Repositories.
	Step 1 (schema) complete. App.config connection string set.
Parallel?	All repository files can be written in parallel after DatabaseHelper exists.

STEP 23 AdminSystem — All Services and Interfaces	
Reason	Admin features (POS, orders, inventory, support, reports) all need their service layer.
	Step 22 complete.
Parallel?	All service files can be written in parallel.

STEP 24 AdminSystem — ViewModels (BaseViewModel first, then all others)	
Reason	All ViewModels inherit BaseViewModel. Write BaseViewModel first, then parallelize the rest.
	Step 23 complete.
Parallel?	After BaseViewModel, all 11 other ViewModels can be written in parallel.

STEP 25 AdminSystem — Views (Login first, then MainWindow, then feature views)	
Reason	Login and MainWindow are the shell. Feature views depend on their ViewModels.
	Step 24 complete.
Parallel?	After Login + MainWindow are ready, all feature views (POS, Orders, Inventory, etc.) can be done in parallel.

STEP 26 CSS and JavaScript — all wwwroot files	
Reason	UI styling and interactivity layer. Can be developed alongside views but tested last.
Depends on	Views exist (Steps 10-18, 25).

	All CSS and JS files are fully independent of each other and can be written in parallel.

STEP 27 End-to-End Testing — Postman collections, manual flow tests for all 16 flowcharts	
Reason	Validate all 16 flows work together: registration, shopping, checkout, voucher, payment, delivery, POS, inventory, support, background jobs, supplier PO.
Depends on	All steps 1-26 complete.
Parallel?	Write Postman collections for Web API endpoints in parallel with finalising CSS/JS.

Taurus Bike Shop · ProjectFilesLibrary v4.0 · 16 Flowcharts · Schema v7.1 · 38 Tables